Indian Institute of Technology (Banaras Hindu University), Varanasi, U.P.

End Semester Examination, November' 2025

Duration:	: 03 Hours	Maximum marks	: 50
Programme	: M.Tech. Mining Engg. (Pt-I) and Ph.D	Semester	: Odd
Course Title	: Reliability Engineering	Course Code	: MN572

Note:

1) *Answer any **FIVE** questions.*
2) *Marks are indicated against each question in the brackets.*
3) *Take* $\Phi(2.055) = 0.98$, $\Phi(2.3879) = 0.99153$ and $X^2_{0.05,\ 3} = 7.81$

Q. 1 (a) After the burn-in period, the lifetime of a mechanical valve is known to be lognormal with t_{med} = 2236 hours and s = 0.41. [5]

(i). Determine the design life of the valve for a mission reliability of 0.98.
(ii). The valve is to be used in a pumping device that requires 5 weeks (840 hours) of continuous use. What is the probability of a pump failure occurring because of the valve failure?
(iii). Compute the Mean Time to Failure of the valve.
(iv). Find the standard deviation of the time to failure of the valve.
(v). Determine the mode of the time to failure of the valve.

(b) A system comprises two identical components, P & Q, operating in parallel and sharing a common load. If P fails, the failure probability of Q increases due to the increased load on Q, and vice versa. Let event A denotes the failure of component P and event B denotes the failure of component Q. If $P(A) = P(B) = 0.05$ and $P(A|B) = P(B|A) = 0.10$. Calculate the reliability of the system. [2]

(c) A gearbox of an LHD has two independent failure modes: Mode–A and Mode–B. Mode–A has a constant failure rate of 0.0003 per hour and Mode–B has an increasing failure rate given by $\lambda(t) = t/(5 \times 10^5)$ per hour. Find the reliability of the gearbox for 100 hours of operation. [3]

Q. 2 (a) A specially designed welding machine has a non-repairable motor with a constant failure rate of 0.05 failures per year. The company has purchased two spare motors. If the designed life of the welding machine is 10 years, what is the probability that two spares will be adequate? [3]

(b) Explain the term 'conditional reliability'. Write down a mathematical expression for the reliability of a component at time t following a burn-in period T_0, when the component has a failure rate $\lambda(t)$. [3]

(c) Write down the expression for the reliability function of the two-parameter Weibull distribution. Define these parameters. Discuss the influence of these parameters on the reliability using suitable plots. [4]

Continued.....

Q. 3 (a) Discuss the redundancy optimization method for maximizing system reliability by allocating active redundancy among the various components within the budget constraints. Assume a serial relationship among a set of independent components within a system. [5]

(b) Bharat Electronics Limited has allocated a budget of ₹70,000 to develop an electronic device comprising three components that must work in series. The following data are available for these three components. Determine the optimum system configuration with redundant units using marginal analysis. Compute the achieved reliability within the allotted budget constraints. [5]

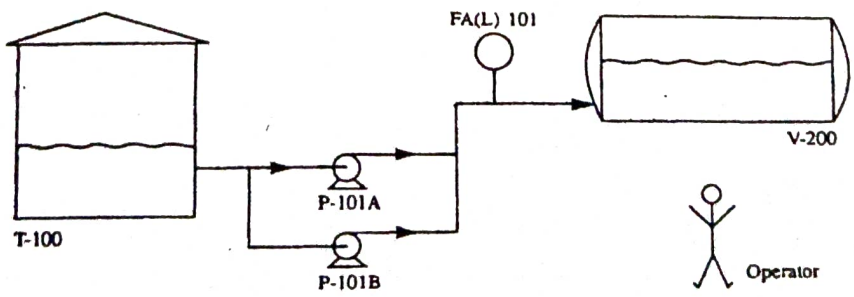
Component	Reliability	Unit cost (₹)
1	0.80	20,000
2	0.90	10,000
3	0.95	7,500

Q.4 (a) Discuss AGREE methods for reliability allocation. [4]

(b) Company XYZ wish to develop a personal computer system that requires a reliability of 0.99 over the first year of use. The developed computer system will consist of three components. The following details of these components are available. Apply the AGREE allocation method to allocate reliability to each of these three components: [6]

Component	Parts count	Importance index	Operating time (hr/yr)
System board	153	0.95	2000
Hard drive	28	0.90	1000
DC power pack	34	1.00	2000

Q.5 A system in which a liquid is pumped from an atmospheric storage tank (T-100) to a pressure vessel (V-200) is shown in figure below. Two pumps are available for this service, P-101A and P-101B. P-101A has a steam turbine driver and P-101B uses an electric motor. Each pump has sufficient capacity to handle the full pumping load by itself. At any given time, one pump is operating, and the other is on standby. If the working pump fails, the operator has 5 minutes to switch to the standby pump. The operator is warned of a no-flow condition by the low-flow alarm FA(L)-101. [10]



Construct a fault tree (FT) of the above system such that the top event is 'system fails to transfer liquid from T-100 to V-200'. Find out the 'Minimal Cut Sets' of the developed fault tree.

Continued.....

Q.6 The failure times (in hours) obtained from a manufacturing unit for testing the filament in electric bulbs are given below. The manufacturer believes that the failures follow an Exponential distribution. Perform a Chi-square test to verify the manufacturer's assumptions. (Take $\alpha = 0.05$). [10]

23	90	765	678	420
556	65	900	59	157
79	454	470	357	273
12	654	129	380	168
67	43	305	259	357
118	234	390	246	568
764	457	540	780	458
456	34	580	400	470

q.7 (a) Explain the following: [5]

(i) Censored data

(ii) Type–I censoring

(iii) Type–II censoring

(iv) Singly censored data

(v) Multiply censored data

(b) Complete the table below. The grouped data reflects failures in operating hours of an air conditioning unit. [5]

i	Time (t_i)	*Number surviving* n_i	*Reliability* $\hat{R}(t_i)$	*Failure density* $\hat{f}(t_i)$	*Hazard rate* $\hat{\lambda}(t_i)$
0	0	44	?	?	?
1	100	41	?	?	?
2	200	36	?	?	?
3	300	28	?	?	?
4	400	18	?	?	?
5	500	6	?		

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M. Tech. Part I/PhDEnd Semester Examination, Session2025-26 Mining Engineering

Subject: Operations Research in Mining, Subject Code: MN561

Time: Three hours **Maximum Marks: 50**

All the symbols have their usual meaning.

Read the question carefully. All question carry equal marks. **Answer any ten questions.**

1.	(a) Discuss the concept of searching for an "optimal solution" in OR, and explain the careful interpretation required in practical management terms. Why is the team approach usually necessary when undertaking a full-fledged OR study, and what diverse skill sets are typically required? (b) Beyond financial savings, what are some of the additional, often more important, non-financial benefits resulting from successful OR studies? Illustrate the use of OR in optimizing logistics and scheduling operations (e.g., airline scheduling, trucking networks). How has OR been applied to critical public services and infrastructure (e.g., police deployment, public health)?	[5]
2.	(a) What are the six major, overlapping phases of a typical Operations Research (OR) study, and how does the mathematical component relate to the overall effort? What are the critical steps and challenges in Defining the Problem and Gathering Data, focusing on objective setting and data quality issues? (b) How is a Mathematical Model formulated, and what are the definitions and roles of its four core components (Decision Variables, Objective Function, Constraints, Parameters)? How do real-world examples illustrate the successful execution of the OR phases, particularly concerning data management, objective conflict, and strategic benefits?	[5]
3.	(a) Define the key terminology used in describing LP solutions, including the distinction between a solution, a feasible solution, the feasible region, an optimal solution. (b) What are the four fundamental assumptions underlying all Linear Programming models? (c) Provide a detailed analysis of the Proportionality assumption, including examples of cost structures that lead to its violation. (d) Provide a detailed analysis of the Additivity assumption, including examples of complementary and competitive interactions between activities that lead to its violation.	[5]
4.	(a) What are Corner-Point Feasible (CPF) solutions, and why does the Simplex Method focus exclusively on them? (b) Define the role of slack variables, and explain the conversion of an inequality-constrained LP model into its augmented form. (c) Algebraically define the properties of a Basic Solution, specifying the relationship	[5]

	between nonbasic variables, basic variables (the basis), and the number of constraints. (d) Detail the algebraic procedure for selecting the Entering Basic Variable (Step 1) and its relationship to the rate of improvement in Z. (e) Explain the Minimum Ratio Test (Step 2) for selecting the Leaving Basic Variable and justify why making a non-minimum choice results in an infeasible solution.	
5.	(a) Define the Transportation Problem (TP) and the Assignment Problem (AP). How are the TP and AP related to each other and to the broader Minimum Cost Flow Problem? (b) Outline the core assumptions (Requirements and Cost) defining the standard Transportation Problem model. (c) When do real-world problems violate the Requirements Assumption, and how are dummy entities used for reformulation? (d) Compare the calculation methodology, computational effort, and expected solution quality for the Northwest Corner Rule (NWCR) and Vogel's Approximation Method (VAM).	[5]
6.	(a) What is the significance of network optimization models in modern Operations Research, and what diverse areas do they impact? (b) Define and differentiate between a path, a directed path, an undirected path, and a cycle in network terminology. (c) Define the relationship concepts of connectivity, spanning tree, and Basic Feasible (BF) solutions in the context of network optimization. (d) Beyond minimizing distance, what are the three major application categories for the Shortest-Path Problem? (e) Summarize the core applications where solving the Minimum Spanning Tree Problem is essential.	[5]
7.	(a) What are PERT and CPM, and what is their combined purpose in project management? What three essential types of information are required to define a project for PERT/CPM analysis? (b) Define the Earliest Start Time (ES) and Earliest Finish Time (EF) for an activity, and state the rule for calculating ES in a forward pass. Define the Latest Start Time (LS) and Latest Finish Time (LF) for an activity, and state the rule for calculating LF in a backward pass. (c) Define Slack for an activity and explain how zero slack identifies Critical Activities. (d) How does the PERT Three-Estimate Approach account for uncertainty in activity durations, and what three estimates are required? Provide the approximate formulas used by PERT to calculate the Mean and Variance of an activity's duration based on the three estimates. What are the three simplifying approximations made by PERT/CPM to estimate the probability of meeting a project deadline ? (e) Define "Crashing" an activity, and explain the role of the time-cost graph and the Normal and Crash points in this analysis. What is the Crashing Problem, and what	[5]

	two primary methods are used to determine the least expensive way to crash activities?	
8.	(a) Define Dynamic Programming, its fundamental purpose, and how it contrasts conceptually with established standard formulations like Linear Programming (LP). (b) What are the eight basic features that characterize problems suitable for formulation and solution using Dynamic Programming procedures?	[5]
9.	The Cost-Less Corp. supplies its four retail outlets from its four plants. The shipping cost per shipment from each plant to each retail outlet is given below.	[5]

		Unit Shipping Cost Retail Outlet			
		1	2	3	4
Plant	1	$500	$600	$400	$200
	2	$200	$900	$100	$300
	3	$300	$400	$200	$100
	4	$200	$100	$300	$200

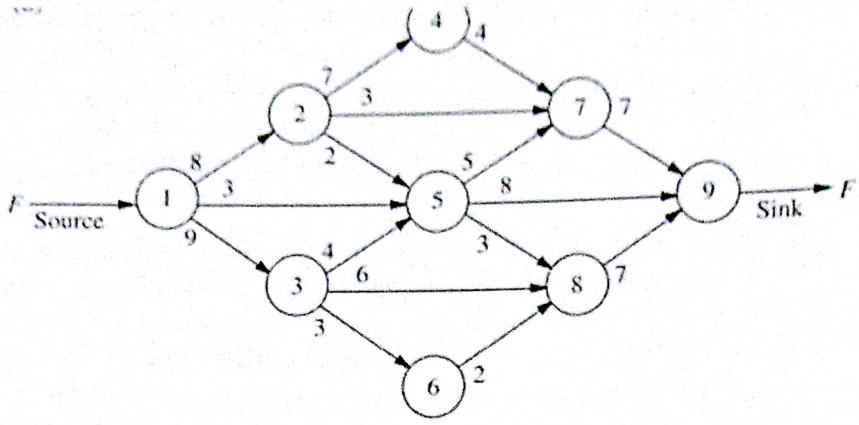
Plants 1, 2, 3, and 4 make 10, 20, 20, and 10 shipments per month, respectively. Retail outlets 1, 2, 3, and 4 need to receive 20, 10, 10, and 20 shipments per month, respectively.

The distribution manager, Randy Smith, now wants to determine the best plan for how many shipments to send from each plant to the respective retail outlets each month. Randy's objective is to minimize the total shipping cost.

(a) Formulate this problem as a transportation problem by constructing the appropriate parameter table.

(b) Use the northwest corner rule to construct an initial BF solution.

10.	(a) What is Queueing Theory and how are its basic components defined? How is the general Birth-and-Death Process (BDP) used to derive steady-state solutions? (b) What is inventory. What are the four critical steps that define the methodology of scientific inventory management? What are the six core components (factors) that determine the profitability of an inventory policy?	[5]
11.	For a given networks , find the flow pattern giving the maximum flow from the source to the sink, given that the arc capacity from node i to node j is the number nearest node i along the arc between these nodes.	[5]

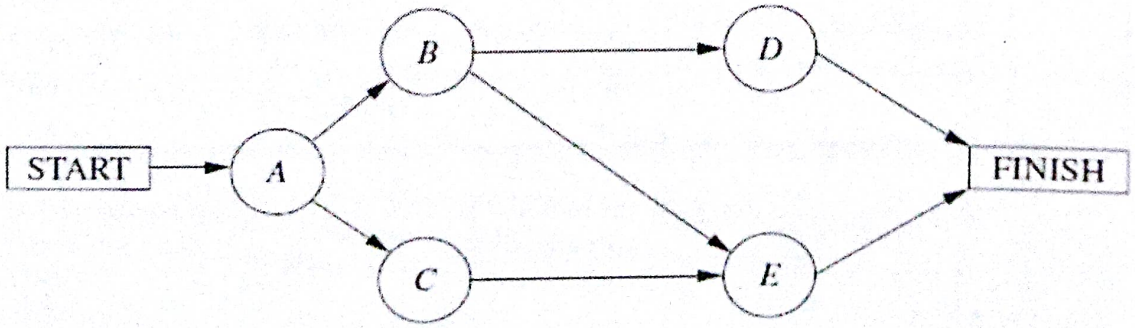


12. You need to take a trip by car to another town that you have never visited before. Therefore, you are studying a map to determine the shortest route to your destination. Depending on which route you choose, there are five other towns (call them A, B, C, D, E) that you might pass through on the way. The map shows the mileage along each road that directly connects two towns without any intervening towns. These numbers are summarized in the following table, where a dash indicates that there is no road directly connecting these two towns without going through any other towns. [5]

	Miles between Adjacent Towns					
Town	**A**	**B**	**C**	**D**	**E**	**Destination**
Origin	40	60	50	—	—	—
A		10	—	70	—	—
B			20	55	40	—
C				—	50	—
D					10	60
E						80

Formulate this problem as a shortest-path problem by drawing a network where nodes represent towns, links represent roads, and numbers indicate the length of each link in miles.

13. Good Homes Construction Company is about to begin the construction of a large new home. The company's President, Ram Manohar, is currently planning the schedule for this project. Ram has identified the five major activities (labeled A, B, . . . , E) that will need to be performed according to the following project network. [5]



He also has gathered the following data about the normal point and crash point for each of these activities.

Activity	Normal Time	Crash Time	Normal Cost	Crash Cost
A	3 weeks	2 weeks	$54,000	$60,000
B	4 weeks	3 weeks	$62,000	$65,000
C	5 weeks	2 weeks	$66,000	$70,000
D	3 weeks	1 week	$40,000	$43,000
E	4 weeks	2 weeks	$75,000	$80,000

These costs reflect the company's direct costs for the material, equipment, and direct labor required to perform the activities. In addition, the company incurs indirect project costs such as supervision and other customary overhead costs, interest charges for capital tied up, and so forth. Ram estimates that these indirect costs run $5,000 per week. He wants to minimize the overall cost of the project. Therefore, to save some of these indirect costs, Ram concludes that he should shorten the project by doing some crashing to the extent that the crashing cost for each additional week saved is less than $5,000.

(a) To prepare for analyzing the effect of crashing, find the earliest times, latest times, and slack for each activity when they are done in the normal way. Also identify the corresponding critical path(s) and project duration.

(b) Use marginal cost analysis to determine which activities should be crashed and by how much to minimize the overall cost of the project. Under this plan, what is the duration and cost of each activity? How much money is saved by doing this crashing?

14. A company is planning the manufacture of a product for March, April, May, and June of next year. The demand quantities are 520, 720, 520, and 620 units, respectively. The company has a steady workforce of 10 employees but can meet fluctuating production needs by hiring and firing temporary workers. The extra costs of hiring and firing a temp in any month are $200 and $400, respectively. A permanent worker produces 12 units per month, and a temporary worker, lacking equal experience, produces 10 units per month. The company can produce more than needed in any month and carry the surplus over to a succeeding month at a holding cost of $50 per unit per month. Develop an optimal hiring/firing policy (only mathematical formulation) over the 4-month planning horizon. [5]

15. An auto company manufactures luxury cars and trucks. The company believes that its most likely customers are high-income women and men. To reach these groups, the company has embarked on an ambitious TV advertising campaign and has decided to purchase one minute commercial spots on two types of programs: comedy shows and football games. Each comedy commercial is seen by 7 million high income women and 2 million high income men. Each football commercial is seen by 2 million high income women and 12 million high income men. A one minute comedy ad costs Rs 50,000, and one minute football ad costs Rs 100,000. Auto company would like the commercial to be seen by at [5]

	least 28 million high income women and 24 million high income men. Determine how auto company can meet its advertising requirements at minimum cost.	
16.	Use the graphical method to solve the problem: Maximize $Z = 2x_1 + x_2$, subject to $x_2 \leq 10$ $2x_1 + 5x_2 \leq 60$ $x_1 + x_2 \leq 18$ $3x_1 + x_2 \leq 44$ and $x_1 \geq 0, \quad x_2 \geq 0.$	[5]

You can Use the following example to explain the answer:

Maximize $Z = 3x_1 + 5x_2$,

subject to the restrictions

$$\begin{aligned} x_1 &\leq 4 \\ 2x_2 &\leq 12 \\ 3x_1 + 2x_2 &\leq 18 \end{aligned}$$

and

$$x_1 \geq 0, \quad x_2 \geq 0.$$

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M.Tech Part I/PhD Mining Engineering, 2025
MN573: Safety & Risk Management in Mines

Time: 3 Hours **Max. Marks: 50**

Note: **Answer any five questions.**

1. Explain the taxonomy of various types of human errors. Discuss with suitable examples, how each of these human errors can be eliminated from the industrial workplaces. (10)

2. Outline the objectives of the 'Risk Management'. Explain the process of risk management. How can it be used as a tool for effective safety management in mines? (10)

3. Discuss the historical developments of various theories for prevention of industrial accidents. Also explain the core concepts/philosophy behind such theories. (10)

4. What do you understand by the term 'motivation'? Explain Maslow's hierarchy of needs, and Expectancy theory of motivation. How do you think these theories can address injury prevention in mines. (10)

5. (a). What is 'value of statistical life'? How can it be utilised as a tool for investment decisions in mines safety? (5)

   (b). Discuss the concept of 'friction period' as used in frictional cost method of accident cost evaluation. How does frictional cost method differ from human capital method? (5)

6. (a). Calculate the expected value and standard deviation of Binomial random variable. (5)

   (b). How is 'safety audit' different from 'safety inspection'? Discuss the process of external safety audit. (5)

7. (a). How will you proceed for chemical hazard management in mines? (5)

   (b). What is fault tree analysis? How can it be helpful as a probabilistic risk assessment tool for identifying the root causes of an accident? (5)

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M.TECH / Ph.D.

COURSE CODE AND NAME: MN-552 – PROJECT MANAGEMENT

End Semester Examination (2025-26 - Odd Semester)

Duration - 3 Hrs

Total Marks - 30

Part A - Detail Answers

Question Number	Question	Marks [5x6]
1.	Explain the importance of Time and Motion Study in improving productivity. Describe the major steps involved in conducting a time study for a repetitive industrial task.	[5]
2.	Discuss the role of Inventory Planning in project management. How do techniques like ABC analysis and Economic Order Quantity (EOQ) help in optimizing inventory costs?	[5]
3.	Compare and contrast tendering and purchasing procedures in the public sector and private sector organizations. Highlight the key differences in transparency and decision-making.	[5]
4.	Explain how Monitoring Techniques and Management Information Systems (MIS) are integrated in project monitoring. Illustrate your answer with a suitable example.	[5]
5.	What are PERT and CPM? Explain the differences between them and discuss how each technique helps in effective project scheduling and control.	[5]
6.	Define Quality Management. Briefly discuss the major concepts, practices, and current trends in quality management that influence modern project execution.	[5]

********** All The Best **********

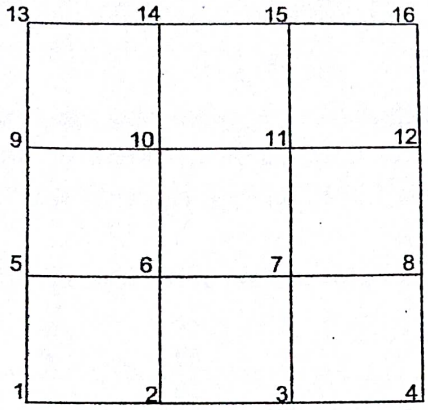
M.Tech. PT. I (SEM. I) and Ph.D., MINING ENGINEERING
SUBJECT – MN 535: FINITE ELEMENT METHODS FOR GEOMECHANICS

End Semester Examination - 2025

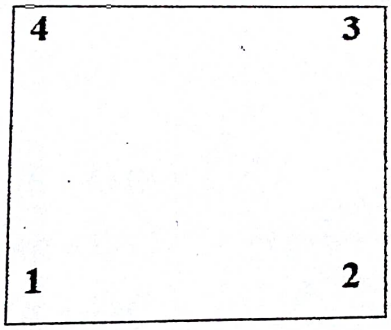
Duration: 3 Hours **Max. Marks: 50**

Note: Answer any five questions. Assume any data if required.

Q. 1 . A 2D structure is discretized into four-point symmetrical elements as shown in figure. The node numbers are assigned in a regular manner. [10]



The stiffness matrix (8 X 8) of an element is shown below.



K =

8	1	0	0	0	0	0	1
	7	0	1	0	0	2	1
		5	2	3	0	0	0
			6	1	0	1	0
				7	2	0	1
					8	0	1
						9	1
							7

Symmetrical

No. 1, 2, 3, and 4 are the local node numbers
The stiffness matrix, shown in the right, is as per the local node numbers

What is the order of the global stiffness matrix of the structure?
Write down the values of the coefficients for row no. 6 of the global stiffness matrix.

Q. 2 A two-dimensional plane strain model is created with linear-elastic and perfectly-plastic model. Write down the step-by-step simulation procedure for updating the material properties after initiating failure state of the element. [10]

Q. 3 Write down the step-by-step modelling procedure for estimating the Local Mine Stiffness (LME) for bord and pillar system of mining. [10]

Q. 4 Derive expression for strain matrix for a constant strain triangular element (3 points) as a function of coordinates of vertices as well as displacements in x and y directions. [10]

Q. 5 The strain matrix for a typical 3-point CST element is calculated considering plane strain condition. [10]

$$\begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{bmatrix} = \begin{bmatrix} 0.004 \\ 0.006 \\ 0.001 \end{bmatrix}$$

Calculate the principal stresses for the same element. Also calculate the factor of safety (FOS) considering material obey Mohr-Coulomb failure criterion. The cohesion and internal frictional angle are 5 MPa and 30° respectively.

Stress-strain relationship for plane strain conditions is expressed as:

$$\begin{bmatrix} \sigma_X \\ \sigma_Y \\ \tau_{XY} \end{bmatrix} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & 0 \\ \nu & 1-\nu & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix} \begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{bmatrix}$$

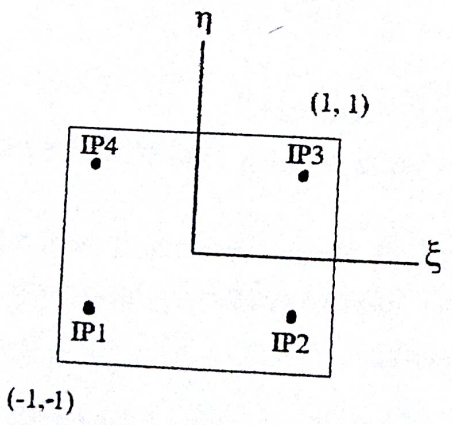
Assume E = 7.2 GPa and $\nu = 0.2$

Q .6 The stiffness matrix in terms of ξ and η for a typical four-point element considering iso-parametric element is expressed as: [10]

$$\begin{bmatrix} A & 0 & 0 & 0 & B & 0 & 0 & 0 \\ B & 0 & 0 & 0 & 0 & 0 & 0 & \\ D & 0 & 0 & D & 0 & 0 & & \\ C & 0 & 0 & 0 & A & & & \\ B & 0 & 0 & 0 & & & & \\ A & C & 0 & & & & & \\ D & 0 & & & & & & \\ C & & & & & & & \end{bmatrix}$$

Symmetrical

$$A = (1+\xi)(1-\eta)$$
$$B = (1-\xi)(1-\eta)$$
$$C = (1+\xi)(1+\eta)$$
$$D = (1-\xi)(1+\eta)$$



	ξ	η
IP1	$\frac{-1}{\sqrt{3}}$	$\frac{-1}{\sqrt{3}}$
IP2	$\frac{1}{\sqrt{3}}$	$\frac{-1}{\sqrt{3}}$
IP3	$\frac{1}{\sqrt{3}}$	$\frac{1}{\sqrt{3}}$
IP4	$\frac{-1}{\sqrt{3}}$	$\frac{1}{\sqrt{3}}$

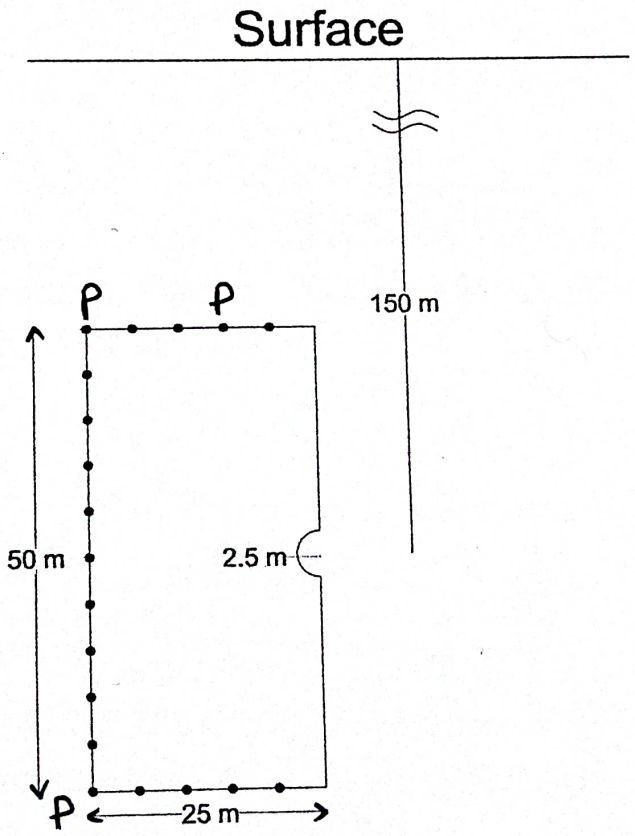
Write down the shape function (N_1, N_2, N_3 and N_4) for the element. Also calculate the stiffness matrix of the element.

Q. 7 An underground tunnel of 5 m diameter is simulated for stress analysis. Half of the tunnel is constructed taking advantage of vertical symmetrical axis as shown in the figure. The boundary is divided into equal length of 5 m. The depth from the center of the tunnel is 150 m. The vertical and horizonal in-situ stress is defined as: [10]

Vertical stress (in MPa) = 0.025 X depth
Horizontal stress (in MPa) = 2.4 + 0.01 X depth

(Where depth is in meters)

Calculate the vertical and horizontal forces to be assigned at various locations (marked as 'P') shown in the model. Also indicate the boundary conditions.



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Indian Institute of Technology (BHU) Varanasi
End-Term Examination (Odd Semester 2025-26)
MME- 306 (Media Environment)

Marks: 60 Time: 3 hrs

Read the instructions carefully before answering.

SECTION-A

Answer all the questions [10]

1. A brand's digital ad delivers a high CTR but also a high bounce rate on the landing page.
   A) The creative message does not align with on-site experience.
   B) The campaign is optimally converting traffic.
   C) The targeting segment is too narrow.
   D) The platform's algorithmic bidding is malfunctioning.

2. A D2C clothing brand invests heavily in Instagram Reels but sees limited sales conversion.
   Which adjustment best aligns with the *See–Think–Do–Care* mindset?
   A) Increase influencer spend for the "See" stage.
   B) Add retargeted Google Shopping ads to move audiences into the "Do" stage.
   C) Reduce storytelling and focus only on discounts.
   D) Shift budget entirely to SEO blogs.

3. During a launch-and-learn phase, a marketer notices that display ads yield high impressions but low ROAS.
   Which next step best demonstrates analytical thinking?
   A) Expand reach before optimising creatives.
   B) Freeze all campaigns until monthly reports arrive.
   C) Use A/B testing to isolate creative vs. placement effects.
   D) Switch budget entirely to email marketing.

4. Africa's progress in *green-hydrogen* production (e.g., Namibia and Morocco) signals a deeper strategic shift.
   Which statement captures it best?
   A) Transition from raw-material exporters to technology-enabled energy partners.
   B) Competing with OPEC in fossil fuels.
   C) Increased dependence on Western subsidies.
   D) Replacement of agriculture with mining industries.

5. Apple's accidental exposure of its App Store front-end code through enabled source maps highlights what core principle of deployment ethics?
   A) Security failures always originate from external attacks.
   B) Configuration discipline is as critical as encryption for safeguarding intellectual property.
   C) UI leaks pose no competitive risk.
   D) Developers cannot influence production security.

SECTION – B

Answer any FIVE of the following questions. [5 x 4]

1. Case: "TechWiz" – A mid-sized IoT firm faces a viral fake-news crisis after reports of product fires spread online. Identify the types of misinformation involved and suggest two verification steps the firm's engineers should take before public communication.
2. Case: "UrbanTech" – A Smart-City project faces privacy backlash due to unclear data policies. Explain two Public-Relations principles UrbanTech failed to apply and recommend steps to rebuild public trust.
3. Case: "Capital Recycling Policy" (NMP / InvIT) – How can opposition parties use agenda-setting and framing to challenge the policy, and what counter-frame would the government employ to accelerate its adoption?
4. Case: "Digital Public Infrastructure (DPI)" – Explain how media framing and agenda-constraining can decelerate adoption of a technically sound policy despite its benefits.
5. Case: "Fake News and Information Disorder" – Discuss how engineers and media professionals can jointly strengthen verification ecosystems to combat misinformation at scale.
6. Using examples from Nokia and Kodak, explain how an engineering mindset can unintentionally create marketing myopia. Suggest one practical step engineers can take to stay customer-focused.
7. Choose any one Brand Archetype (The Hero, The Creator, or The Everyman). Describe how a brand can express this archetype through its communication style and visual identity.
8. Digital Marketing Scenario: A start-up runs a campaign that reaches one million users but records a low conversion rate. Using the See–Think–Do–Care framework, identify two likely reasons and propose one corrective action.

SECTION – C

Answer any THREE of the following questions. [3 x 10]

1. Discuss how gatekeeping and linkage shape the interaction between engineering systems and public perception during policy crises. Illustrate with examples from class.
2. In light of the *Fake News and Information Disorder* framework, analyse the long-term implications of misinformation for engineers and society. Suggest how media literacy can be integrated into professional practice.
3. In the age of AI-generated content, how can brands preserve authenticity and human connection while leveraging automation in marketing and storytelling? Discuss using examples of at least two brands that balance technology efficiency with emotional resonance.
4. You are the Brand Head for a new Indian fintech startup. Using insights from *Brand Archetypes* and *Foundations of Brand Management, and popular culture* explain:

   a) Which archetype combination would help build long-term trust and aspiration among Gen Z users, and
   b) How digital storytelling can reinforce this brand purpose consistently across platforms.
5. Popular culture increasingly influences user behaviour in digital and built environments. Explain how culturally informed design and communication approaches can improve inclusivity and system effectiveness.